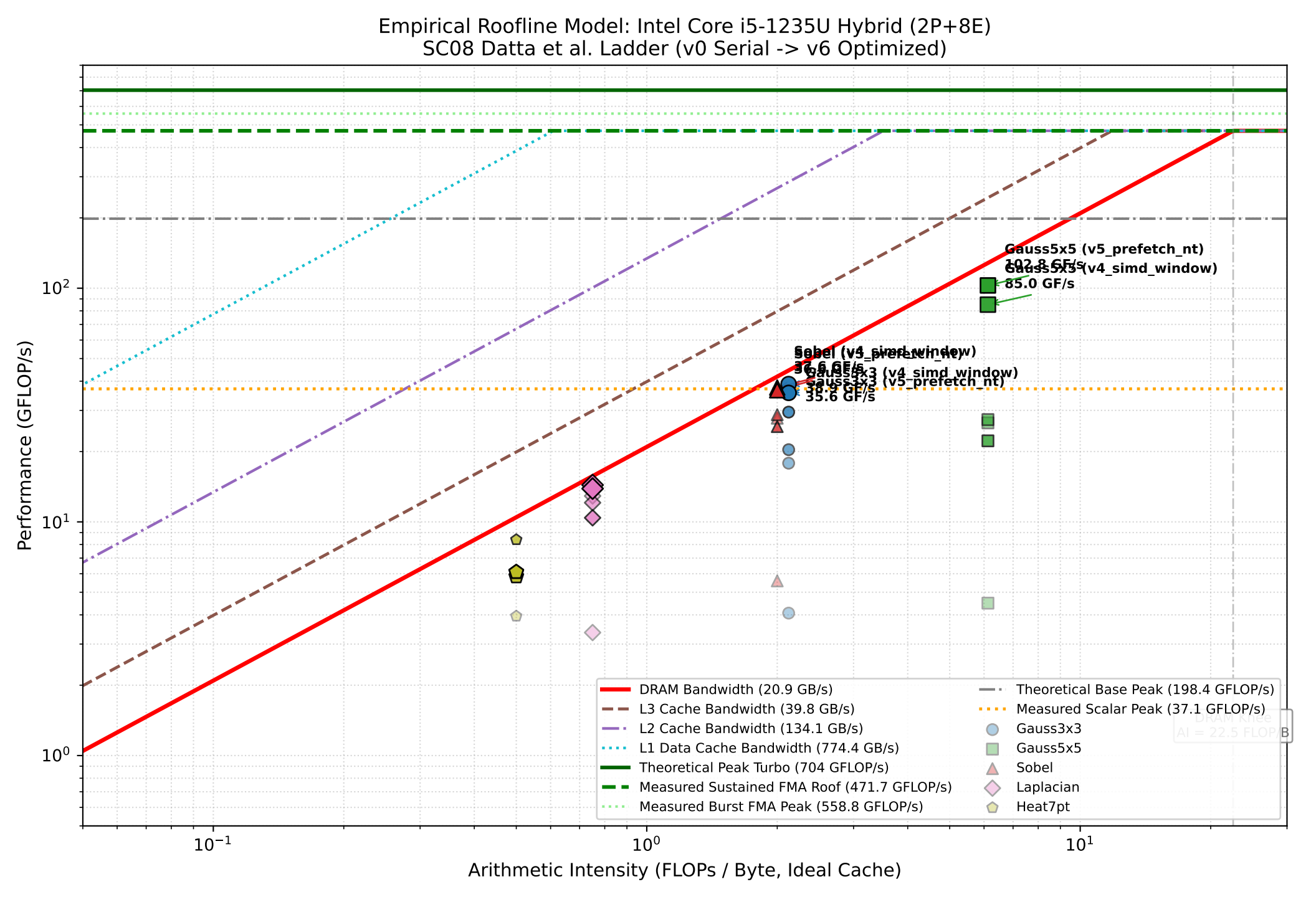
Empirical Roofline Model: Intel Core i5-1235U Hybrid (2P+8E)
SC08 Datta et al. Ladder (v0 Serial -> v6 Optimized)
Performance (GFLOP/s)
Arithmetic Intensity (FLOPs / Byte, Ideal Cache)
Gauss5x5 (v5_prefetch_nt)
102.8 GF/s
Gauss5x5 (v4_simd_window)
85.0 GF/s
Sobel (v4_simd_window)
Sobel (v5_prefetch_nt)
Gauss3x3 (v4_simd_window)
Gauss3x3 (v5_prefetch_nt)
35.6 GF/s
DRAM Bandwidth (20.9 GB/s)
L3 Cache Bandwidth (39.8 GB/s)
L2 Cache Bandwidth (134.1 GB/s)
L1 Data Cache Bandwidth (774.4 GB/s)
Theoretical Peak Turbo (704 GFLOP/s)
Measured Sustained FMA Roof (471.7 GFLOP/s)
Measured Burst FMA Peak (558.8 GFLOP/s)
Theoretical Base Peak (198.4 GFLOP/s)
Measured Scalar Peak (37.1 GFLOP/s)
Gauss3x3
Gauss5x5
Sobel
Laplacian
Heat7pt